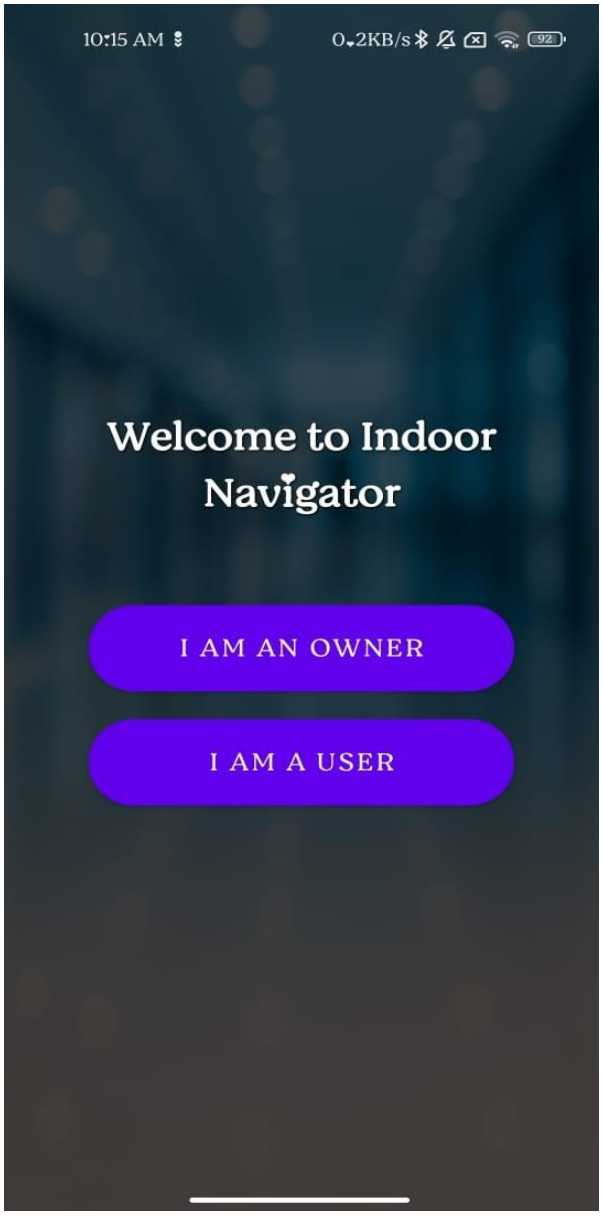
Indoor Navigation System Documentation

Feature: User Role Selection



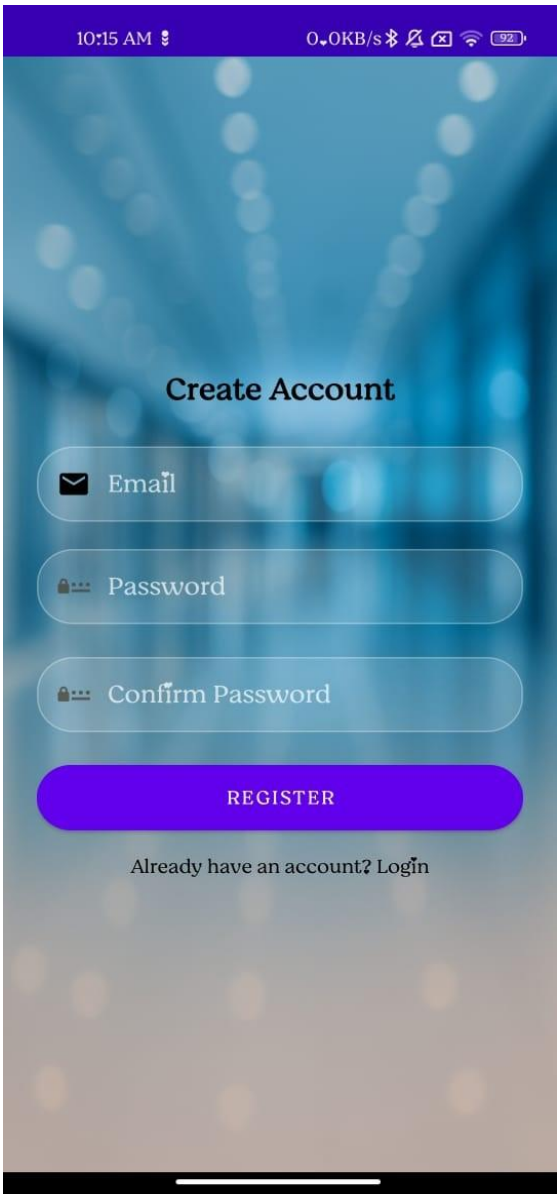
Technical Explanation:
Developed using Kotlin to facilitate role-based navigation and UI branching. It serves as the landing module for identity verification before activity redirection. The interface uses Material components to ensure a standard Android design language. Logic handles secure transitions between Admin and User dashboard perspectives.

Technology Used:
Kotlin, Android Material UI Components.

Purpose: *To differentiate user roles within the application. It securely verifies identity and then directs individuals to the correct interface ensuring that Admins and Users each access their respective dashboards with appropriate permissions.*

Indoor Navigation System Documentation

Feature: Admin Account Registration



Technical Explanation:
Utilizes Firebase Authentication to create new administrative credentials for building management. Kotlin validation check confirms password complexity and account status during registration. It asynchronously pushes user profiles to the cloud, ensuring global accessibility.

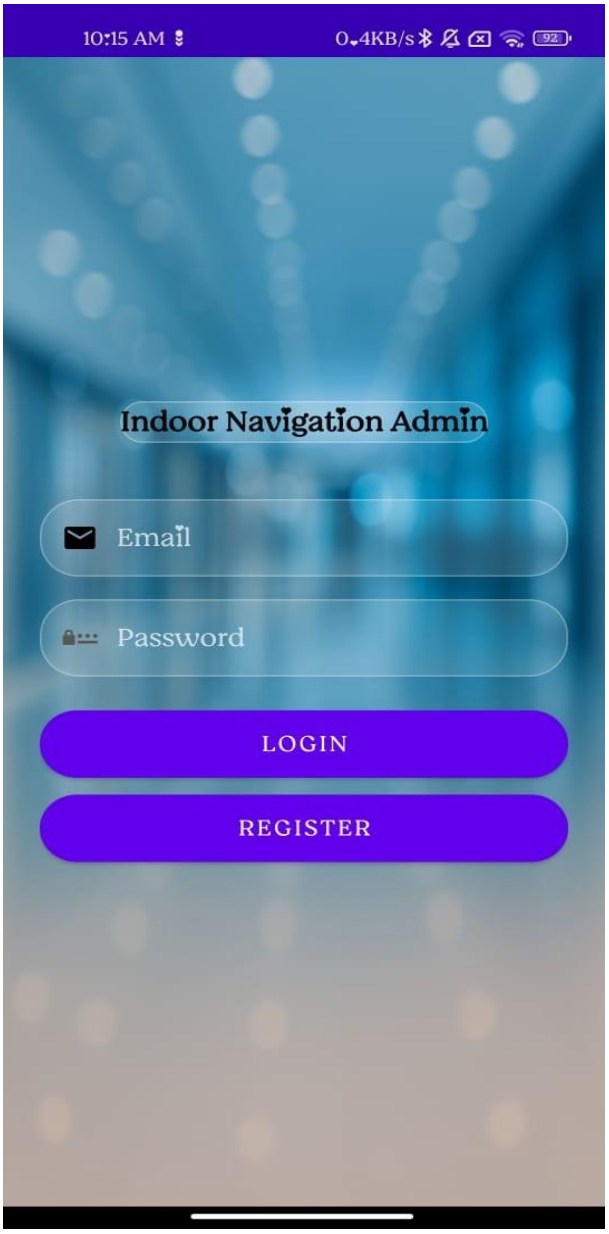
Technology Used:
Firebase Authentication, Kotlin

Purpose:

Onboards new administrators to manage their buildings' indoor navigation paths securely.

Indoor Navigation System Documentation

Feature: Secure Admin Login



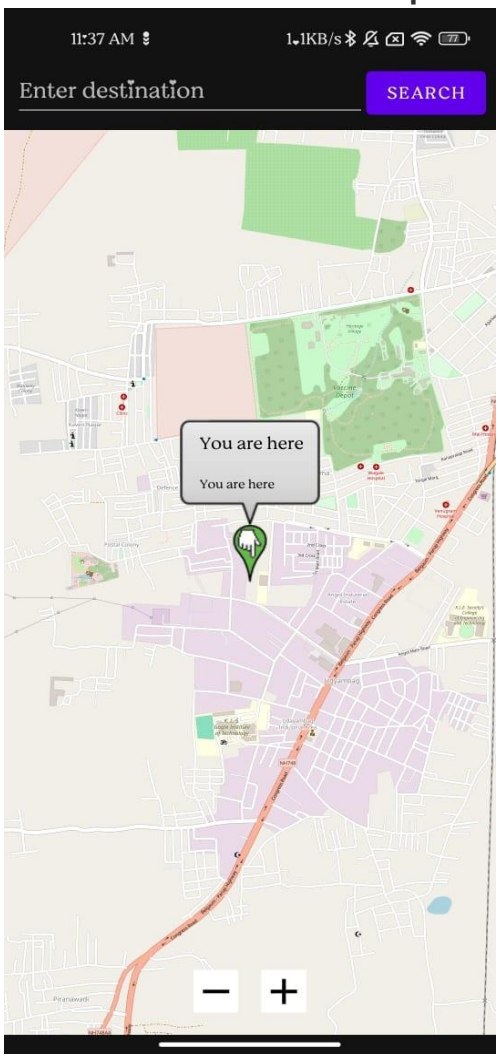
Technical Explanation:
Implements Firebase Authentication to manage secure identity access for building administrators. Built with Kotlin, it handles real-time input validation and asynchronous login requests. It ensures that only authorized accounts can modify indoor floor plan layouts

Technology Used:
Firebase Authentication, Kotlin.

Purpose:
Restricts management features and private building data to authenticated architectural owners.

Indoor Navigation System Documentation

Feature: Global Map Orientation & Site Metadata Setup



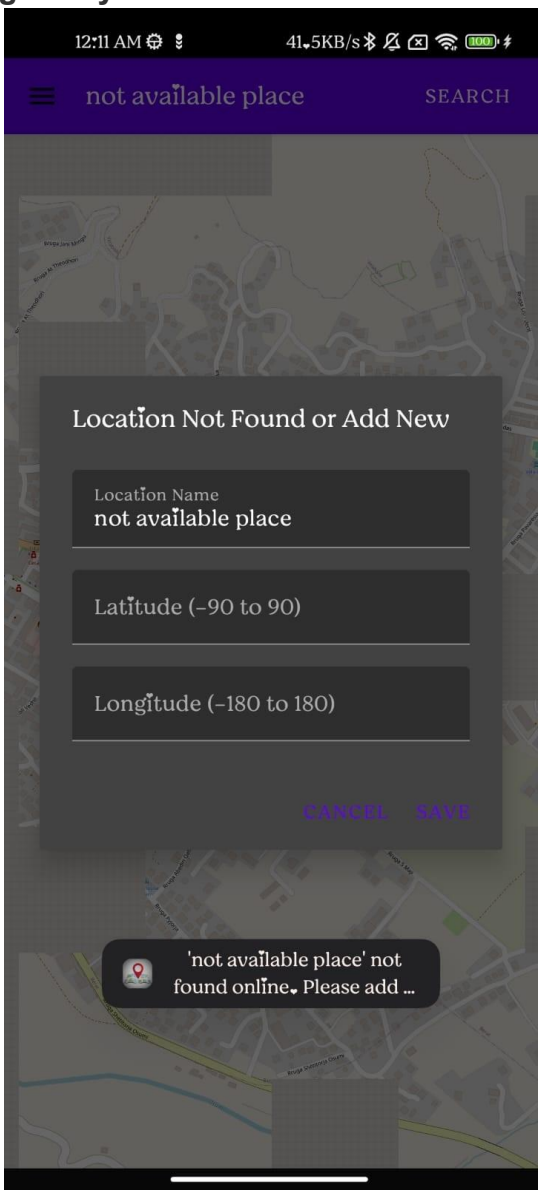
Technical Explanation:
Integrates the osmdroid library to display open-source map tiles with low-latency rendering. Kotlin logic retrieves real-time GPS data via the Location Provider to pin user coordinates. It uses custom marker overlays to visualize geographical distance from the target building.
Uses Kotlin to pair physical building coordinates with specific indoor floor plan images. The module saves location names and geocoordinate types into the Firebase database. It acts as the primary configuration screen for defining new landmark profiles.

Technology Used:
osmdroid (OpenStreetMap rendering), Android Location Provider (GPS), Kotlin, Firebase Database (location + coordinates storage).

Purpose:
Orients users geographically on a world map before they enter a specific indoor facility.
Defines the specific landmark and geographical data for a site before indoor mapping begins.

Indoor Navigation System Documentation

Feature: Manual Geocoding Utility



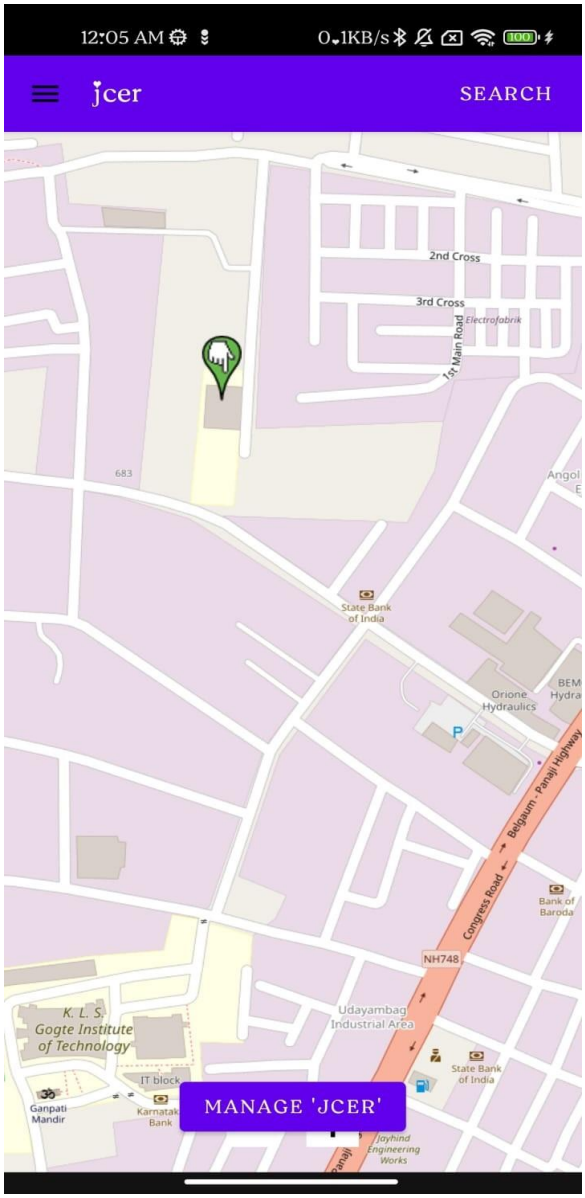
Technical Explanation:
Provides a Kotlin-based fallback for adding custom locations not indexed in public map APIs. It saves latitude and longitude values directly to Firebase to create new geofences. This interface manages floating-point precision for coordinates to ensure accuracy.

Technology Used:
Kotlin, Firebase Database

Purpose:
Allows admins to register unmapped coordinates for new or specific indoor navigation zones.

Indoor Navigation System Documentation

Feature: Owner’s Central Dashboard



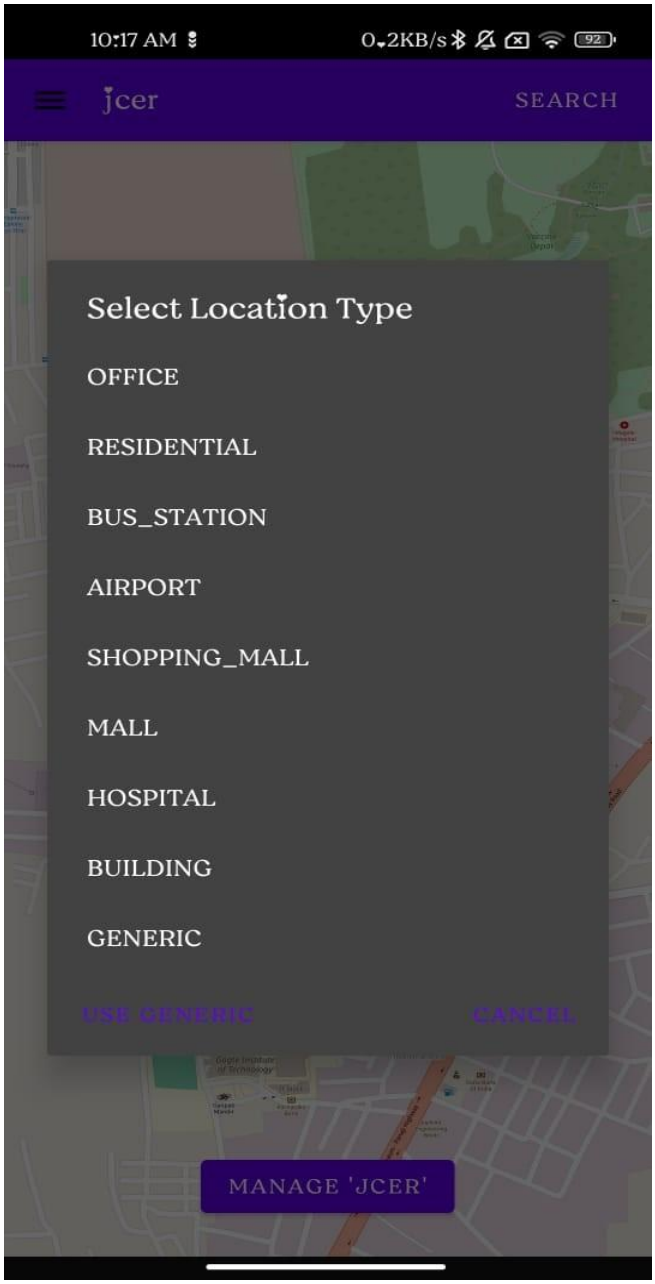
Technical Explanation: This screen uses the OSMDroid library to display an OpenStreetMap-based world map with custom markers. The location data for these markers is fetched in real-time from a Firebase Firestore database. The "Manage" button navigates to the editor screen (AddDetailsActivity).

Technology Used: OSMDroid, Firebase Firestore, Firebase Authentication, RecyclerView.

Purpose: *To provide the owner with a central dashboard to view all their created building locations on a world map. It acts as the main portal for editing existing indoor maps.*

Indoor Navigation System Documentation

Feature: Building Classification



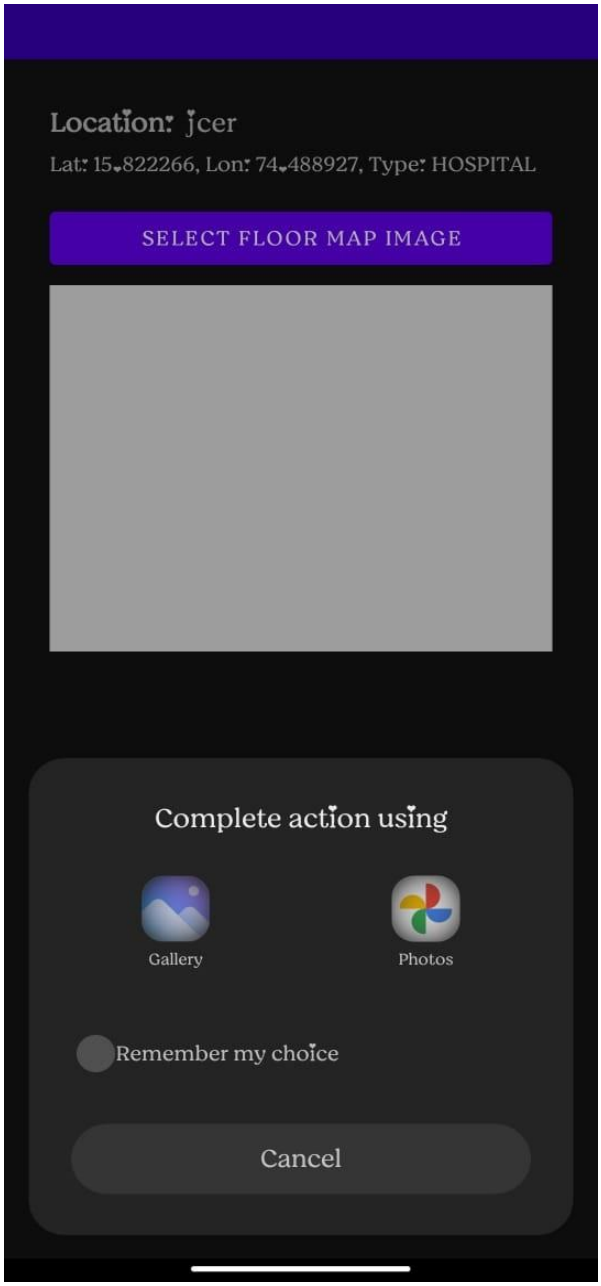
Technical Explanation:
Uses a Kotlin-based selection dialog to categorize building utilities like Hospitals or Malls. This data is recorded in Firebase to optimize search algorithms and context-aware navigation. It guides the ML Kit extraction logic on what specific types of room labels to expect.

Technology Used:
Kotlin Dialog, UI, Firebase Database

Purpose:
Categorizes building environments to help users find and navigate specific types of facilities.

Indoor Navigation System Documentation

Feature: Blueprint Asset Selection



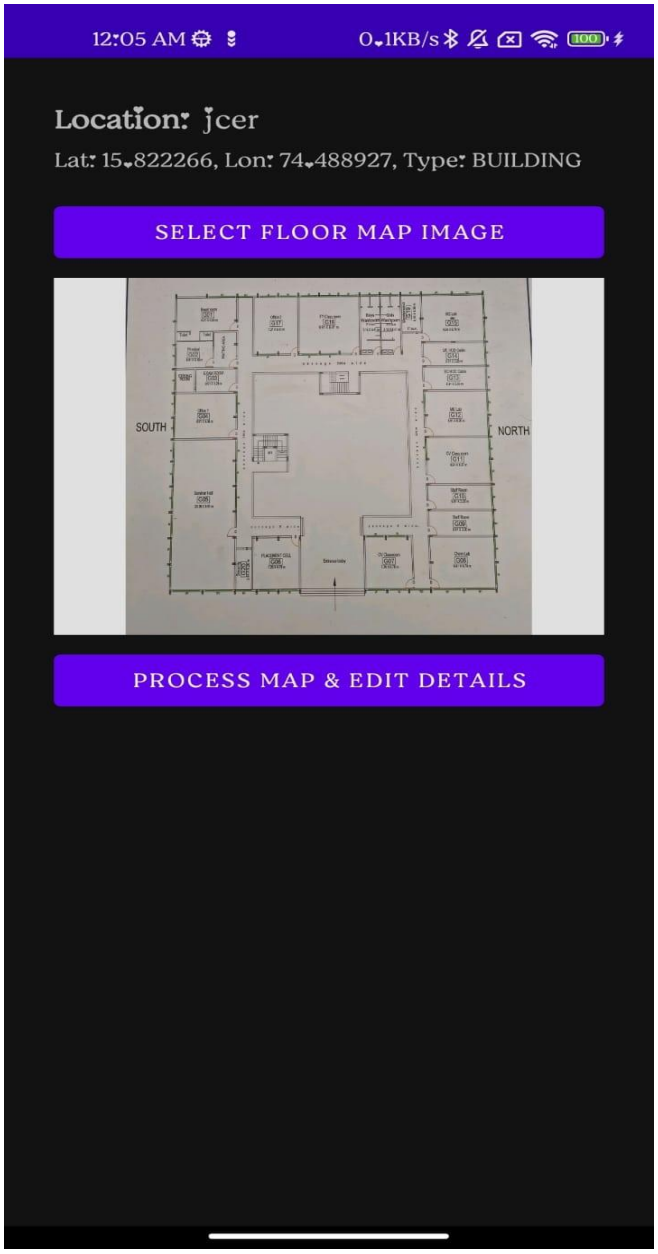
Technical Explanation:
Implements the Android Storage Access Framework to fetch blueprint images from the gallery. Kotlin manages URI permissions for processing high-resolution PNG or JPEG floor plan files. verified image formats are passed to the map processing environment.

Technology Used:
Android Storage Access Framework, Kotlin (URI handling)

Purpose:
Enables administrators to upload the base architectural image for digital maps.

Indoor Navigation System Documentation

Feature: Map Processing Screen



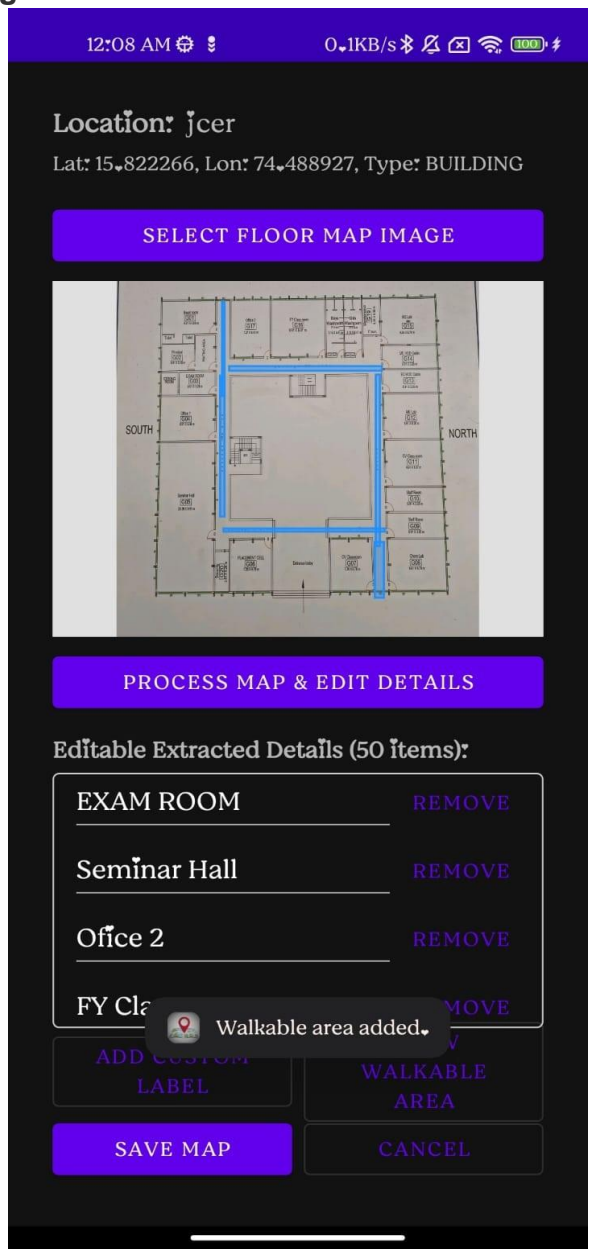
Technical Explanation: This activity uses an ActivityResultLauncher to select a floor plan image from the device's gallery. Google's ML Kit is then used to perform on-device OCR to extract text from the image. The results are displayed in a RecyclerView for the owner to edit.

Technology Used: Android SDK, ActivityResultLauncher, Google ML Kit (Text Recognition), SubsamplingScaleImageView, RecyclerView.

Purpose: *To allow a building owner to upload a floor plan image. It then automatically extracts room labels to digitize the map for navigation*.

Indoor Navigation System Documentation

Feature: Intelligent Map Digitization



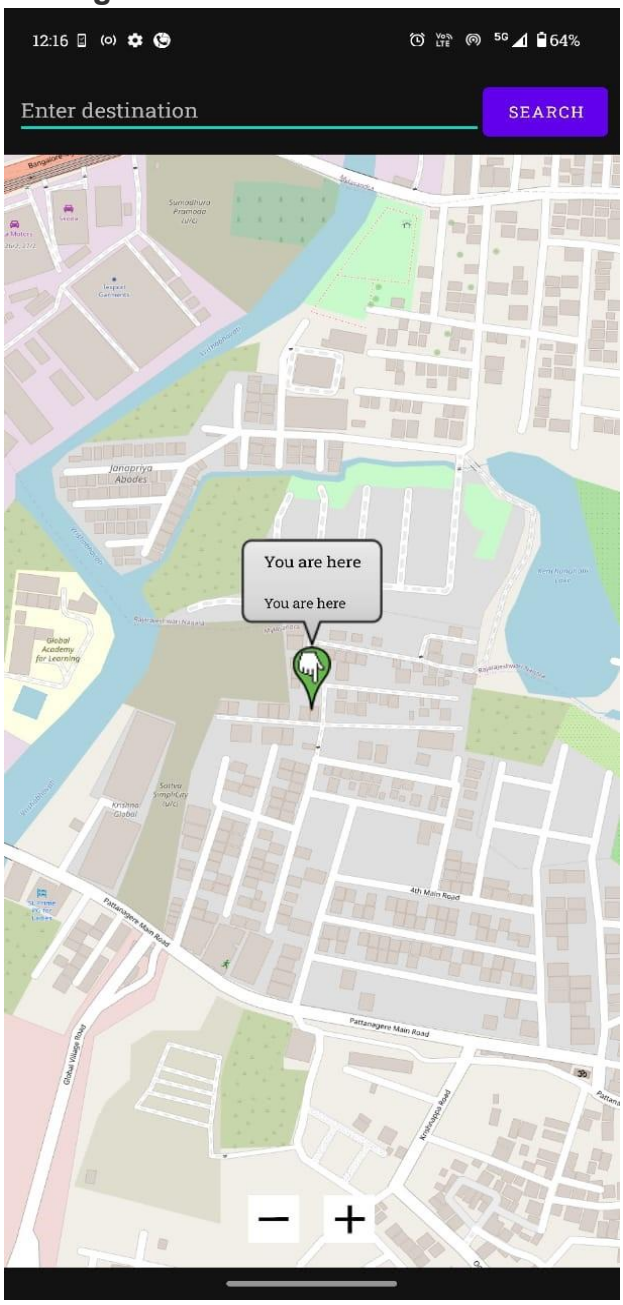
Technical Explanation:
Leverages ML Kit for text recognition to automatically extract room names from the floor plan image. Kotlin tools allow admins to define walkable areas over the static blueprint. Resulting navigational graphs and coordinates are stored in the Firebase database.

Technology Used:
Google ML Kit (OCR Text Recognition), Kotlin, Firebase Database

Purpose:
Transforms static floor plan images into interactive maps with searchable rooms and paths.

Indoor Navigation System Documentation

Feature: Live Outdoor Positioning



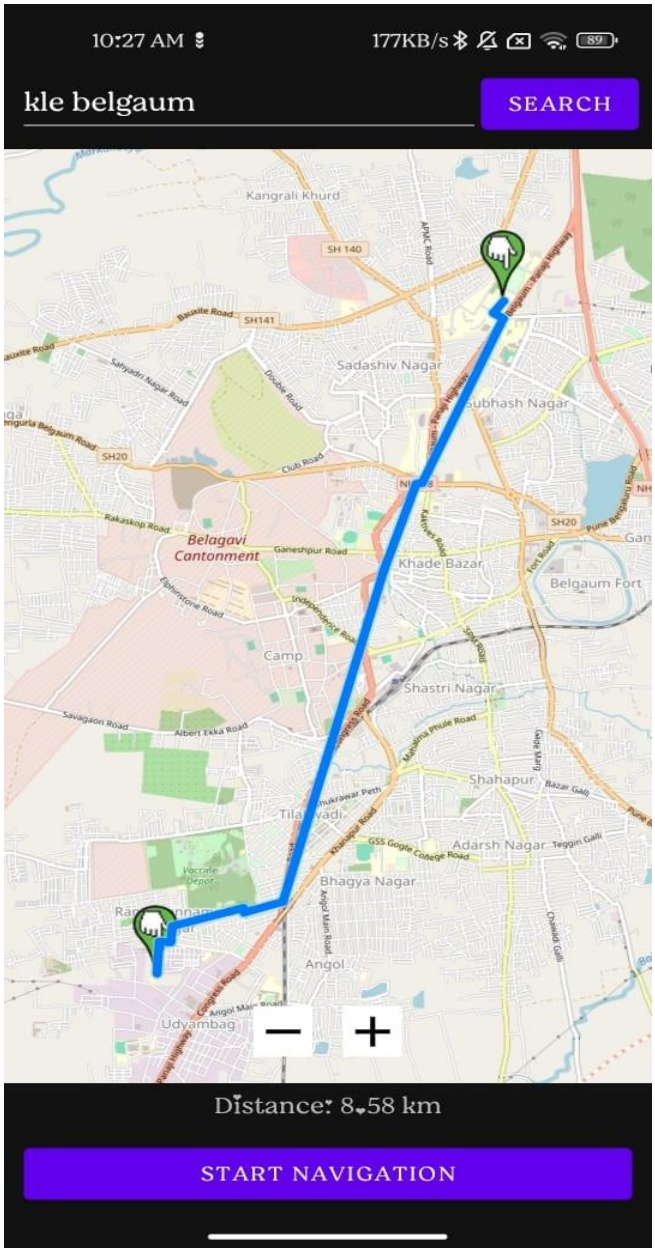
Technical Explanation:
Integrates osmdroid to provide a real-time global positioning interface. Kotlin handles the location provider callbacks to update the 'You are here' marker dynamically. OpenStreetMap tiles are cached for efficient map rendering and panning.

Technology Used:
osmdroid, GPS Location Provider, Kotlin

Purpose:
Identifies the user's current external coordinates before starting the final campus navigation.

Indoor Navigation System Documentation

Feature: Outdoor Route Calculation



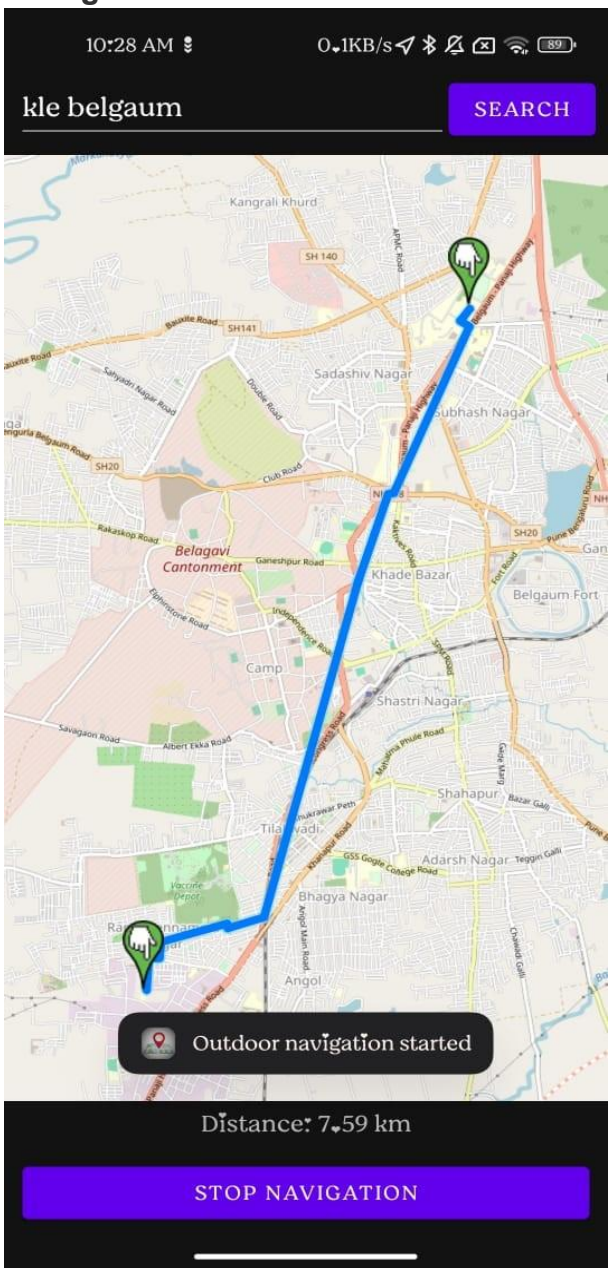
Technical Explanation:
Uses a routing API integrated with osmdroid to calculate the optimal path between two geocoordinates. Kotlin performs distance calculations and renders the polyline on the map. It bridges the gap between the start point and the destination building entrance.

Technology Used:
Routing API, osmdroid Polyline Rendering, Kotlin

Purpose:
Visualizes the external travel distance and route required to reach the physical building location.

Indoor Navigation System Documentation

Feature: Active Outdoor Tracking



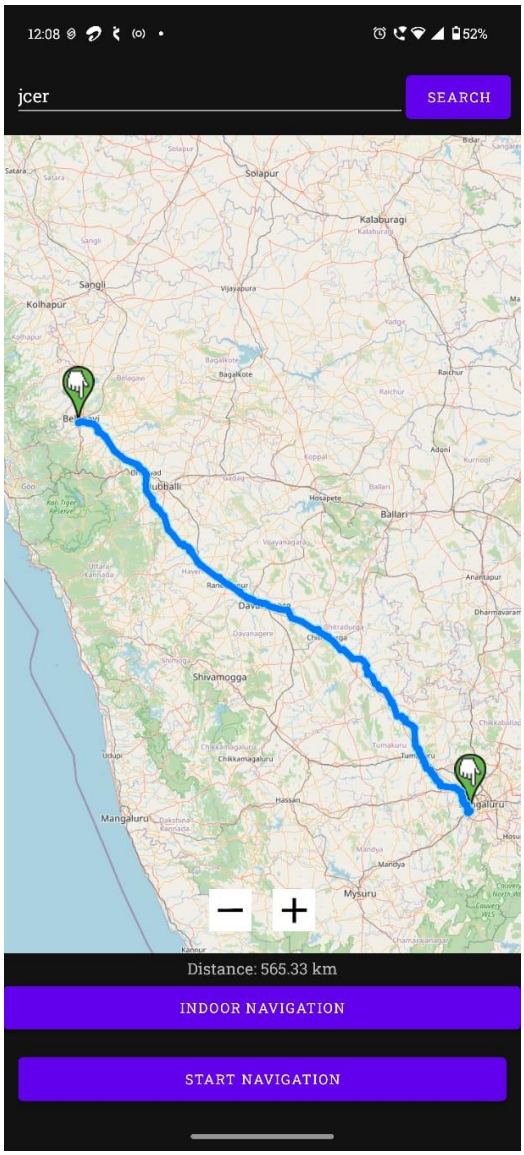
Technical Explanation:
Executes a foreground service in Kotlin to track user progress along the calculated outdoor route. Real-time GPS updates are compared against the osmdroid polyline for path adherence. Features a dynamic 'Stop Navigation' toggle to manage location listeners.

Technology Used:
Android Foreground Service GPS Updates, osmdroid, Kotlin

Purpose:
Provides real-time guidance and distance feedback during the approach to a target building.

Indoor Navigation System Documentation

Feature: Entry Point to Indoor Navigation



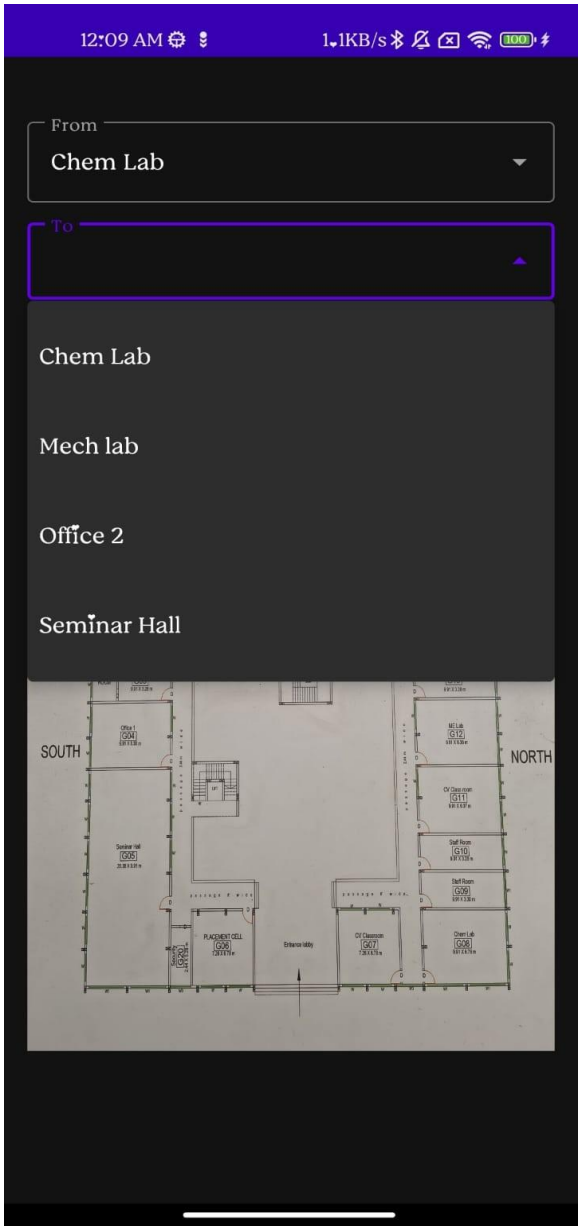
Technical Explanation: Clicking the INDOOR NAVIGATION button fetches the specific indoor map document from Firebase Firestore, which includes the floor plan image, all OCR labels, and the owner-drawn walkable areas. Clicking the START NAVIGATION button. It activates a live, turn-by-turn navigation mode directly on the existing OSMDroid map. This requires using a routing engine (like OSRM) to provide real-time location updates and directional guidance on the OSMDroid map overlay, guiding the user to the building's entrance without ever leaving your application.

Technology Used:
OSMDroid, a third-party routing engine (e.g., OSRM), Firebase Firestore.

Purpose:
To guide a user from their current outdoor location to the main entrance of the target building. It acts as the handoff point from standard outdoor navigation to the specialized indoor navigation.

Indoor Navigation System Documentation

Feature: Indoor Room Selection



Technical Explanation:
Fetches a dynamic list of rooms from Firebase Realtime Database based on the current building ID. Implements a Kotlin Searchable Dropdown for selecting 'From' and 'To' waypoints. Data is retrieved asynchronously to ensure a non-blocking UI experience.

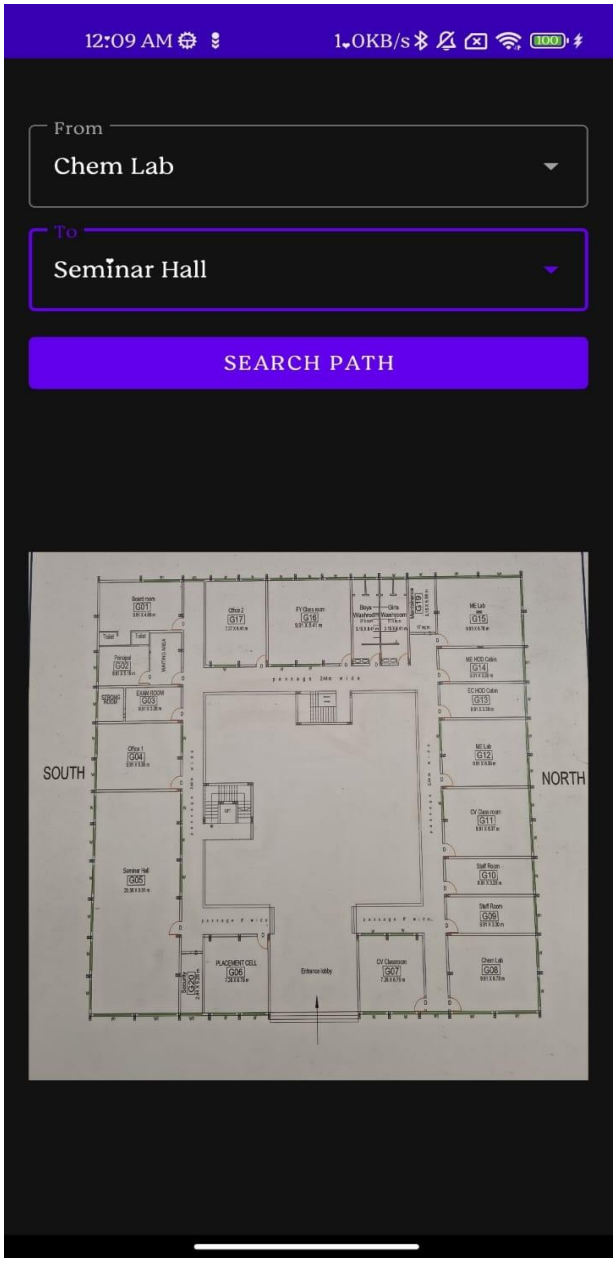
Technology Used:
Firebase Realtime Database, Searchable Dropdown (AutoComplete UI), Kotlin

Purpose:

Allows users to choose specific start and end points within the digitized internal floor plan.

Indoor Navigation System Documentation

Feature: Indoor Path Request



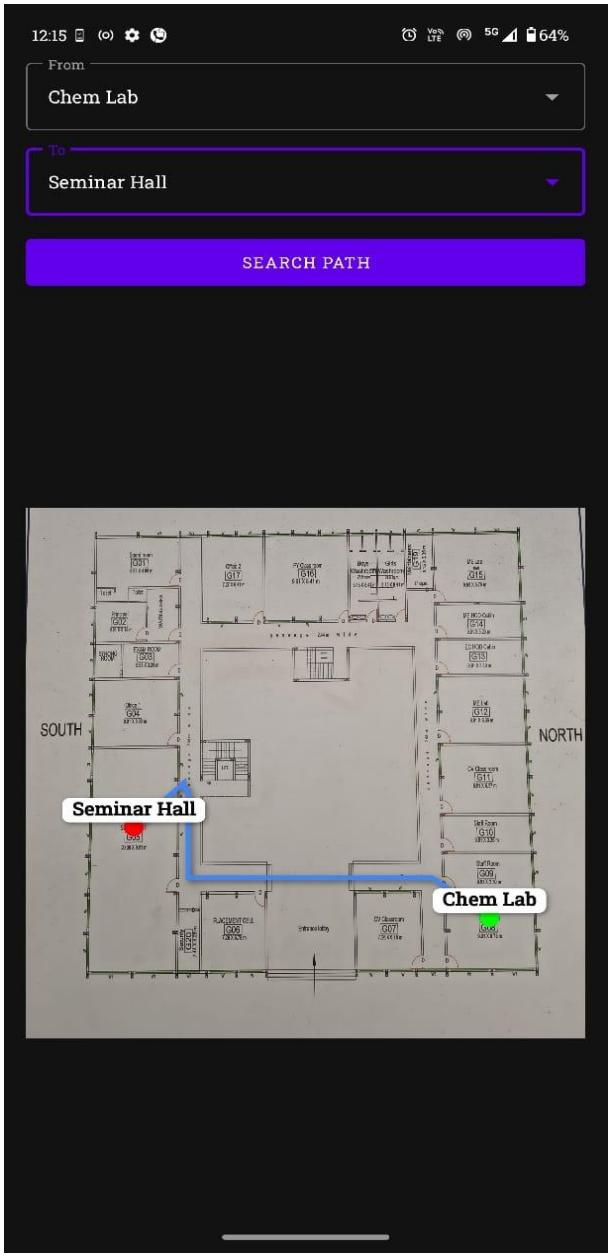
Technical Explanation:
Links selected room identifiers to their localized pixel coordinates stored in Firebase. Kotlin logic validates the selection to ensure both points are present on the current floor plan image. Prepares the spatial data for the internal pathfinding algorithm execution.

Technology Used:
Firebase Database (pixel coordinates), Kotlin

Purpose:
Sets the navigation parameters for generating a specific indoor route between rooms.

Indoor Navigation System Documentation

Feature: Indoor Path Visualization



Technical Explanation:
Utilizes a custom Kotlin Canvas overlay to draw the calculated path directly onto the floor plan image. It maps logical nodes to image-space coordinates for visual alignment. Shortest-path logic calculates the walkable route based on digitized labels.

Technology Used:
Custom Android Canvas Drawing, Shortest Path Algorithm (A*), Kotlin

Purpose:
Displays the final step-by-step visual path for navigating between rooms inside the building.